

CSE 4849

Human Computer Interaction (HCI)

Lecture 8: Applying ethnography in design

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Overview

- Ethnography – Review
- Structuring ethnographic presentation: a framework
- Contextual Design
- Contextual Inquiry
- Work modeling
- Affinity diagram

Ethnography

- A *qualitative* description of the *human social condition*, based on *fieldwork* and *observation*.
 - Qualitative implies a lack of statistical evidence, but a richer experiential understanding of gathered data.
 - Social phenomenon occur within a culture, and exist when there is interaction between individuals.
 - “The field” is where work is done (i.e., the context of the work).
- *The goals of design and the goals of ethnography are at opposite ends of a spectrum*

Structuring ethnographic presentation

- A framework to structure the ethnographic presentation in a way that enables designers to use them. Three main dimensions of this framework are:
 - The distributed co-ordination dimension
 - Focuses on the distributed nature of the tasks and activities, and the means and mechanisms by which they are coordinated.
 - The plans and procedures dimension
 - Focuses on the organizational support for the work, such as workflow models and organizational charts, and how these are used to support the work.
 - The awareness of work dimension
 - Focuses on how people keep themselves aware of others' work. No-one works in isolation, and it has been shown that being aware of others' actions and work activities can be crucial element of doing a good job.
 - Implications here relate to the sharing of information.

Contextual Design

- Contextual Design
 - Developed by Hugh Beyer and Karen Holtzblatt in 1998
 - Developed to handle data collection and analysis from fieldwork for developing a software-based product
 - It provides a structured approach for gathering and representing information from fieldwork such as ethnography with the purpose of feeding it into design.
 - Has 7 parts
 - Contextual Inquiry
 - Work Modeling
 - Consolidation
 - Work Redesign
 - User Environment Design
 - Mockup and Test with Customers
 - Putting It into Practice

Contextual Inquiry

- Contextual Inquiry
 - An approach to ethnography study
 - The designers works as an apprentice to the users
 - Contextual interview
 - at users' workplace (workstation)
 - 2 to 3 hours long
 - Combination of observation, discussion, and reconstruction of past events.
 - 4 main principles:
 - Context — see workplace & what happens
 - Partnership — user and developer collaborate
 - Interpretation — observations interpreted by user and developer together
 - Focus — project focus to help understand what to look for

Contextual Design

- Work modeling
 - Flow model
 - the interactions, the activities and the relationships among people involved in the work
 - Sequence model
 - The steps to accomplish an activity
 - Sequences are collected during the contextual interview
 - Artifact model
 - The products used in the context, the documents used and produced
 - Cultural model
 - The norms (written and implicit), the roles of the people
 - Constraints on the system caused by organizational culture
 - Physical model
 - The environment (e.g. the layout of the office), the physical spaces

Work flow model

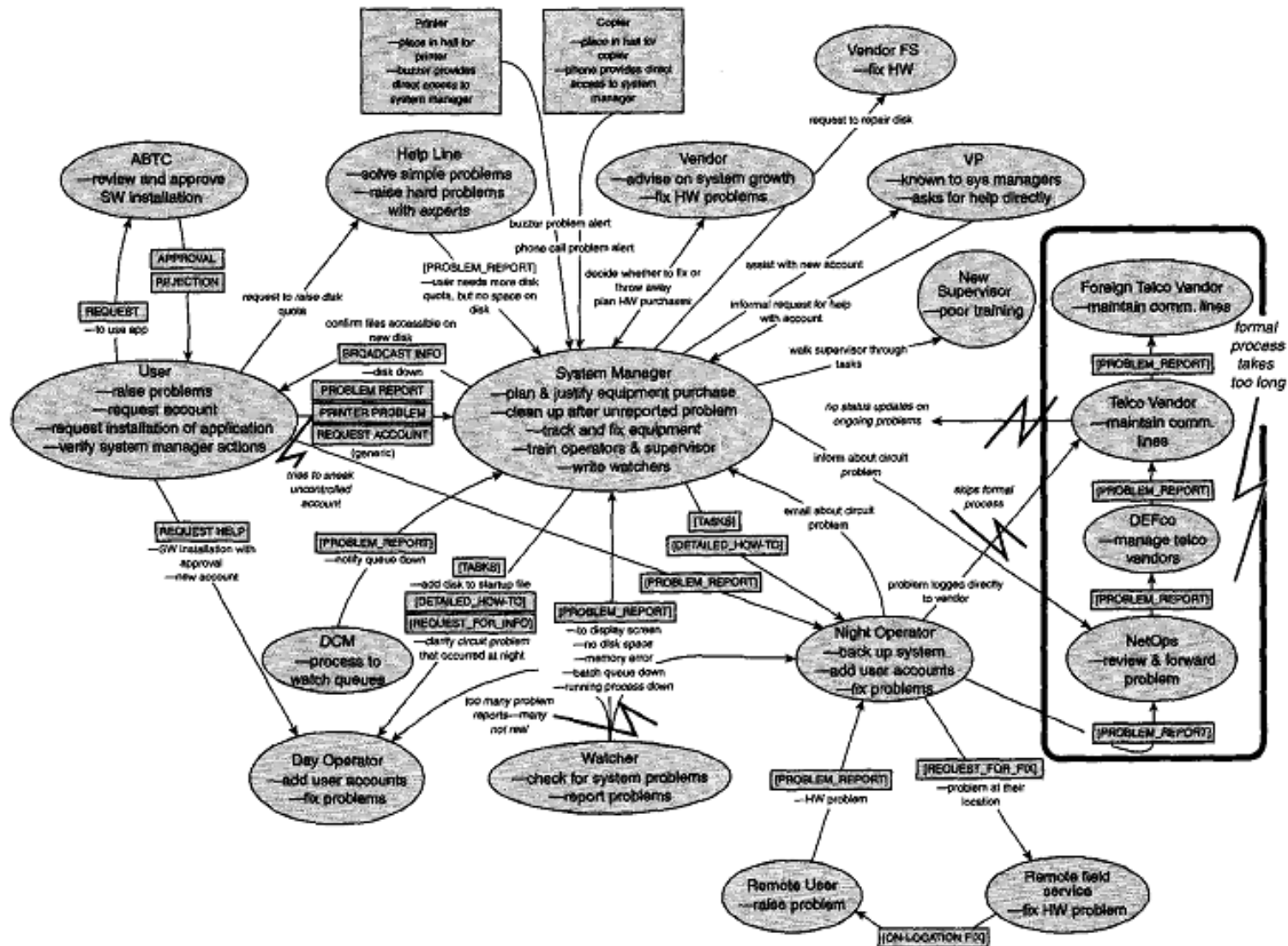


Figure 9.5 An example work flow model.

Each bubble in the flow model represents a role. A job title plays multiple roles. Behind each role is a list of responsibilities the role plays. This flow model shows a process with the key work groups coordinating at each step in the process. Both formal and informal activity is depicted.

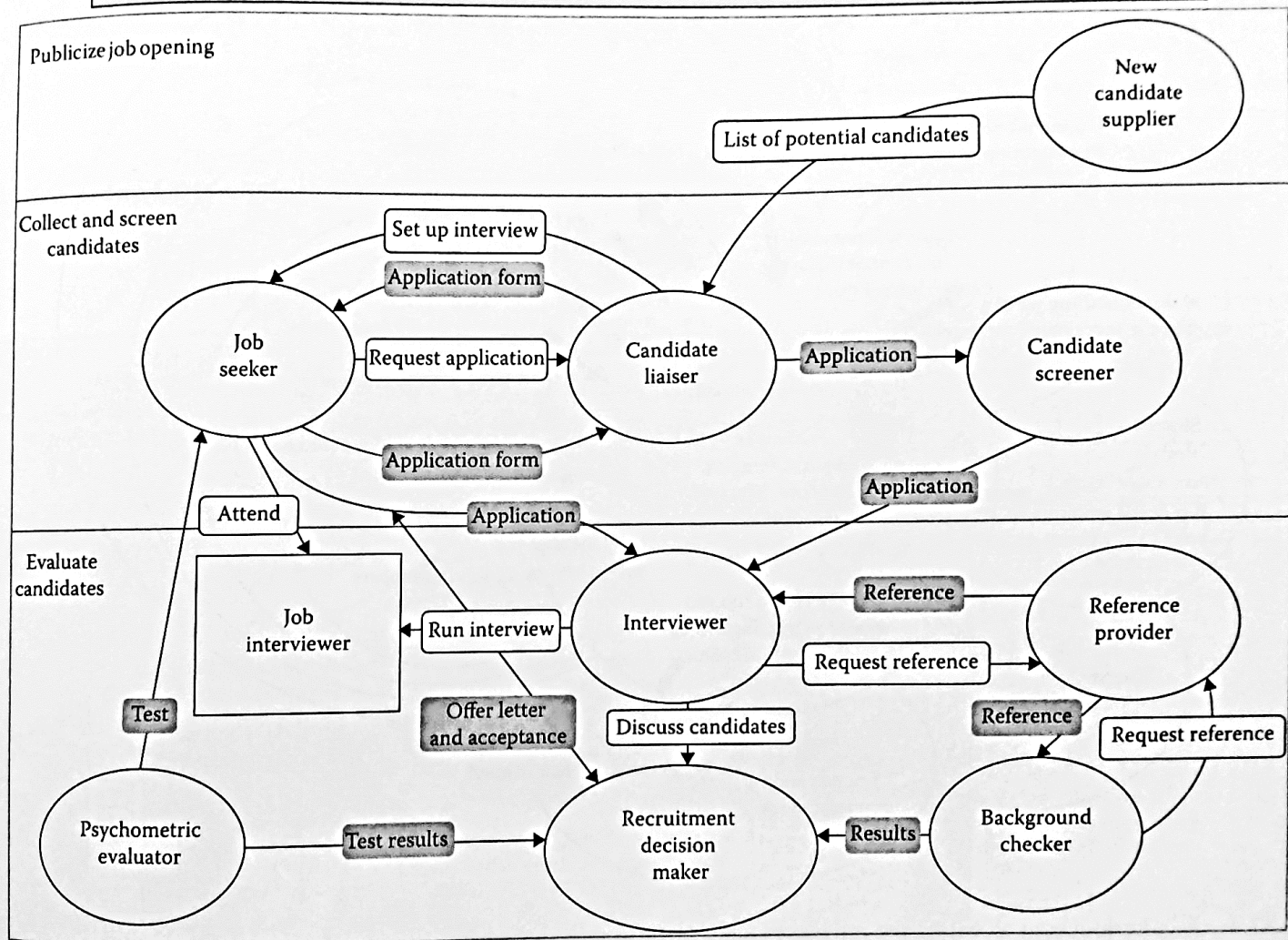


FIGURE 43.2 A flow model revealing the hiring process.

Sequential model

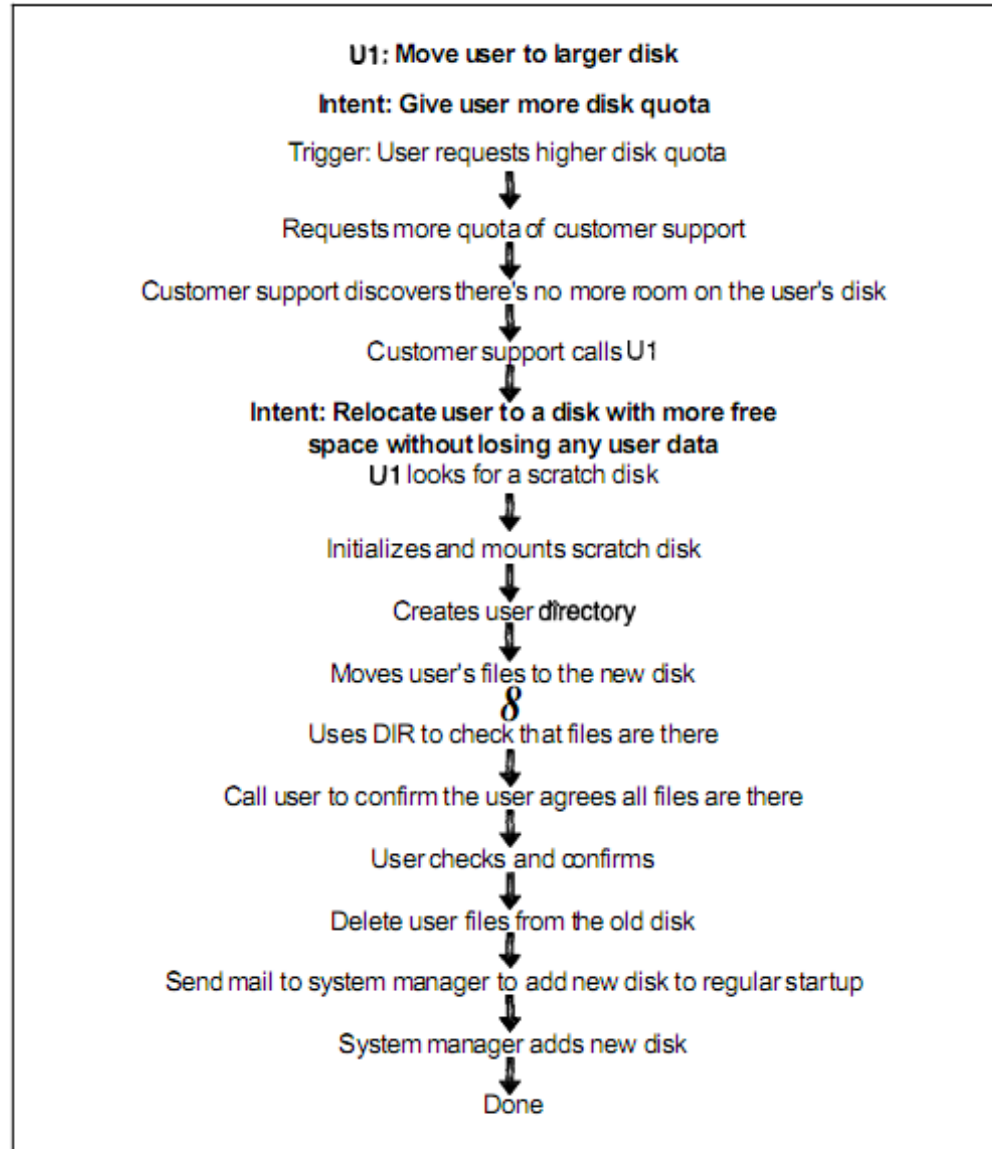


Figure 9.6 An example sequence model.

Cultural model

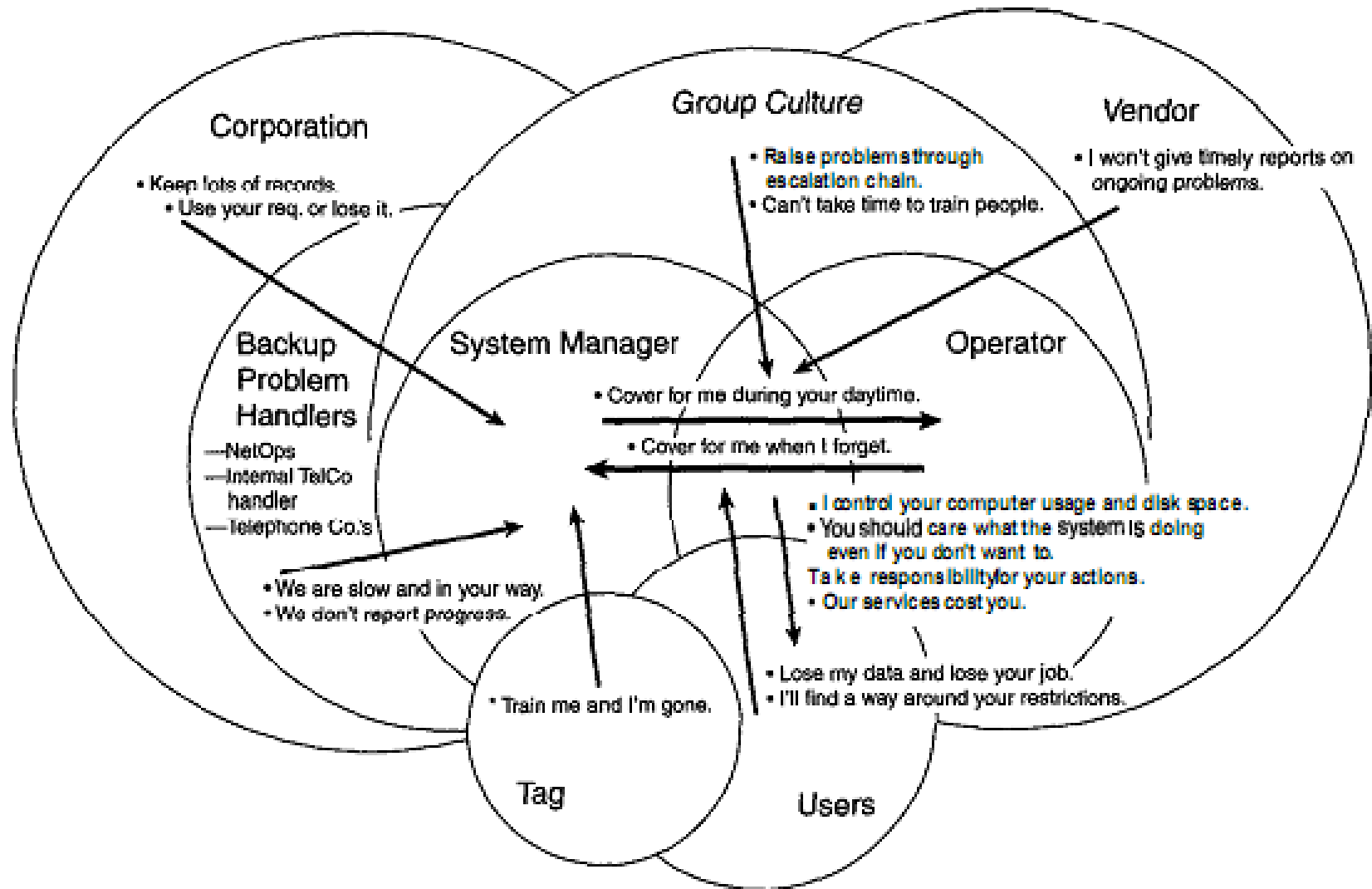


Figure 9.7 An example cultural model.

Physical model

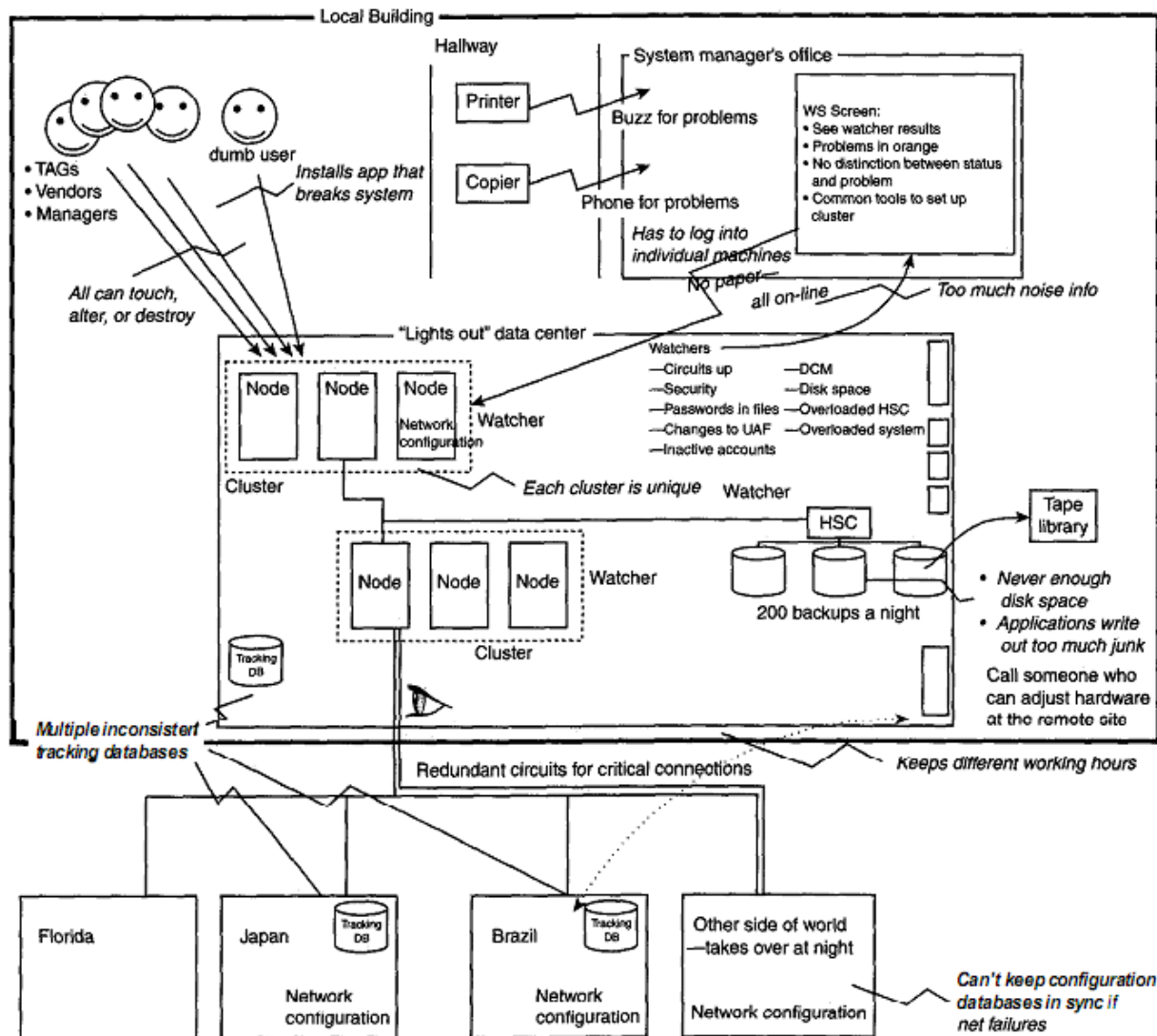


Figure 9.8 An example physical model.

Consolidation

- Consolidation
 - The models produced by the different teammates are shared and reconciled by teamwork
 - This part is also important for reducing the impact of biases
- Consolidating the models
 - Each contextual inquiry (one for each user/developer pair) results in a set of models
 - These need to be consolidated into one view of ‘the work’
 - Affinity diagram
 - Organizes interpretation session notes into common structures and themes
 - Categories arise from the data
 - Diagram is built through induction

The Affinity Diagram

- The affinity diagram organizes the individual interpretation session, or affinity, notes into a wall-sized, hierarchical diagram grouping the data into key issues under labels that reveal the customer's needs.
- The affinity shows in one place the common issues, themes, and scope of the customer problems and needs.
- The affinity acts as the voice of the customer and the issues it reveals become the basis for user requirements.
- It is your fastest and best method to see all the issues across your user population.

The Affinity Diagram

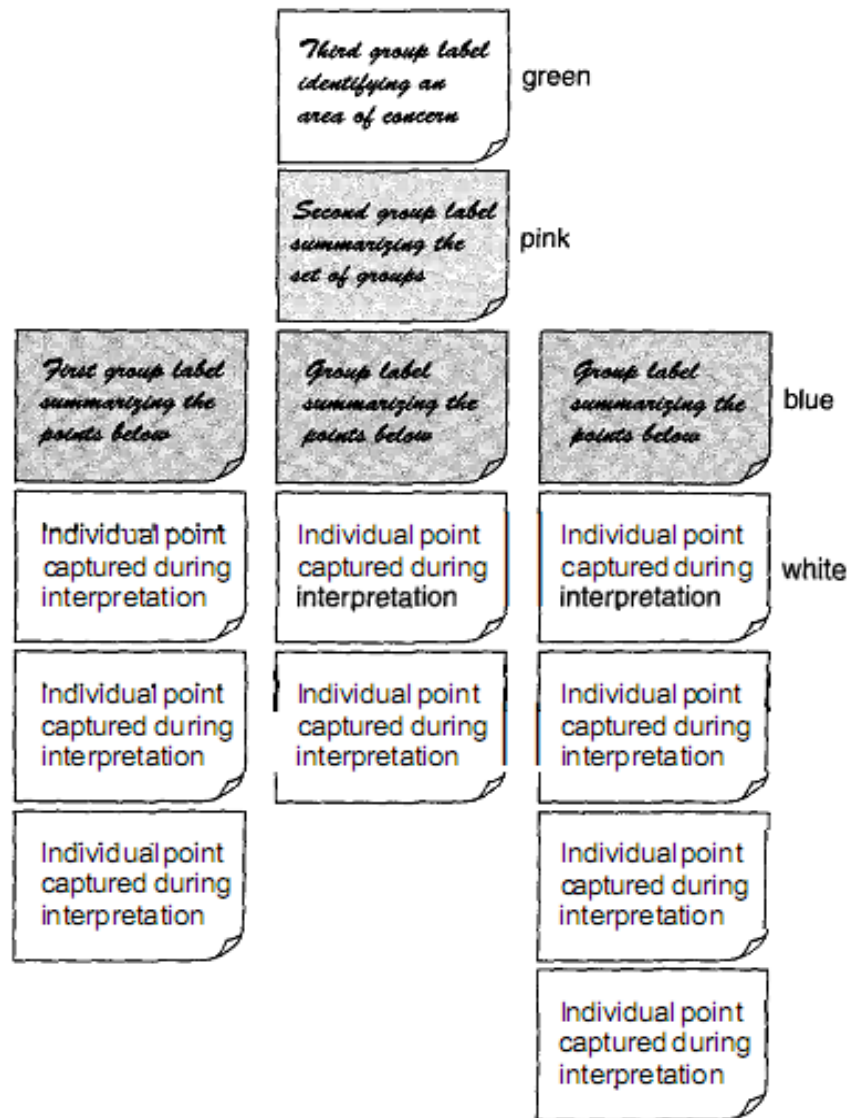
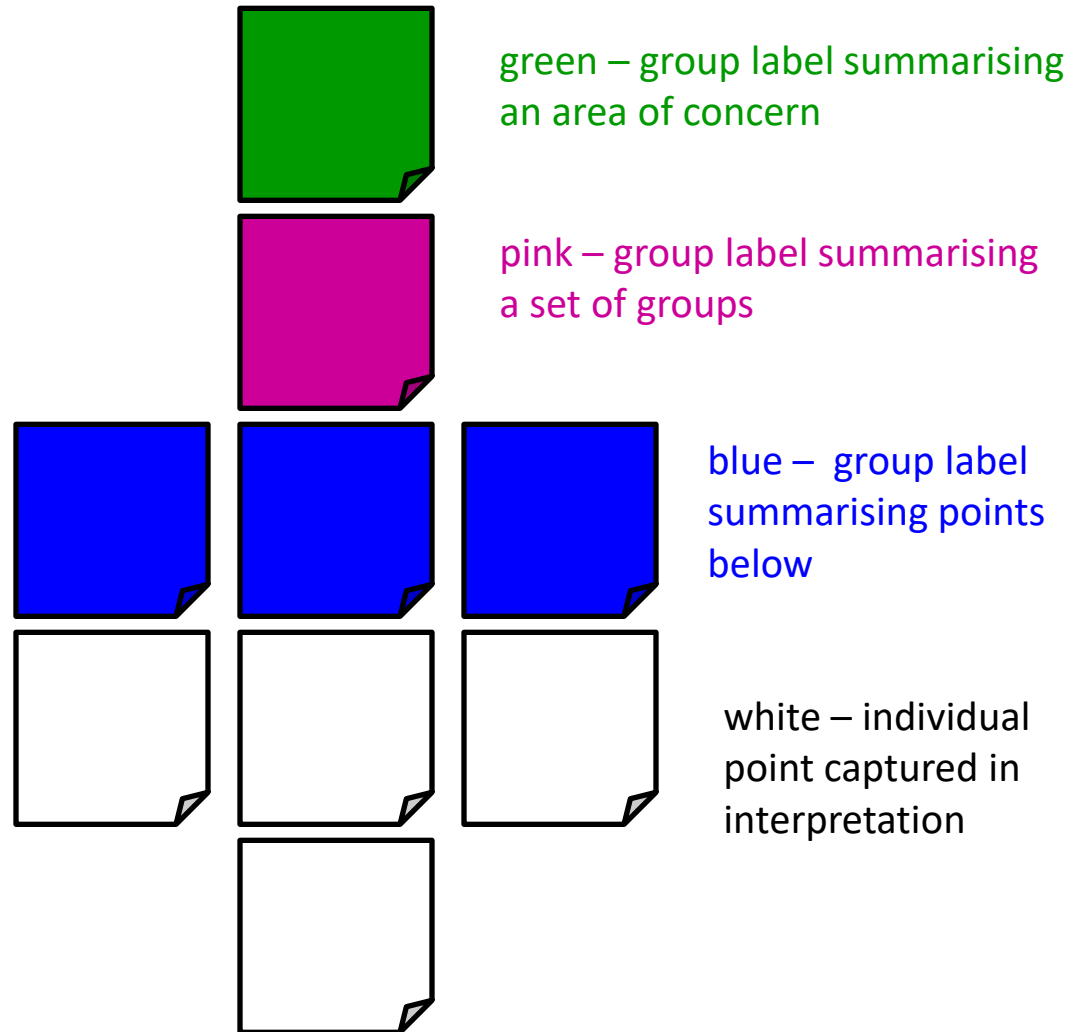


Figure 9.9 The structure of an affinity diagram.

The Affinity Diagram

- Think about a single application.
- White- Write wish list one-in-one page. What you want to do with the system. Use WH- questions.
 - What I want to do?
 - How I want to do it?
 - Why I want to do?
 - Who else want to do?
 - Where I want do?
 - When I want to do?
- Blue – Summarize the white points to form group label
- Pink – Group the blue summaries
- Green – The area of concern you want to design and implement.



Homework

- *A contextual inquiry journey in a pizza shop*
- Have a pizza in your favorite pizza shop and do some field work
 - Describe people, environments and interactions
 - Use the 5 models of CD
- Prepare for consolidation exercise
 - Affinity diagram exercise

- Resources
 - HCI Handbook chapter 49: Contextual Design
 - HCI Handbook chapter 50: An Ethnographic Approach to Design